INCH-POUND
MIL-F-24402/2(SH)
3 April 1990

MILITARY SPECIFICATION SHEET

FILTER HOUSINGS, HYDRAULIC, DUPLEX

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-F-24402.

CLASSIFICATION. The dual bowl filter housings covered by this specification sheet shall be designated using part or identifying numbers (PIN) as follows:

M	24402	/2	N	L	-WXYZ
Prefix to indicate an item defined by inch-pound units	Specification number	Applicable specification sheet	Type	Porting	Indicator form

Examples: M24402/2ND-M90 is a dual bowl housing with selector valve, no bypass relief, left and right hand outlet ports, and mechanical pop-up indicators which actuate at 90 lb/in^2 differential.

M24402/2RS is a dual bowl housing with selector valve, bypass relief, standard right hand outlet port, and no differential pressure indicators.

FILTER ASSEMBLY TYPE. The filter assembly type shall be designated by one of the following letters to indicate whether or not a bypass relief is installed:

N - No bypass relief
R - Bypass relief installed

PORTING. The filter assembly outlet porting shall be designated by one of the following letters to indicate the location of the outlet port.

S - Standard right hand outlet (see figure 1)
L - Left hand outlet (see figure 1)

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D - Dual outlet (both left and right outlets drilled). Unless otherwise specified (see figure 1), assembly shall be furnished with blind flange installed on left hand outlet.

DIFFERENTIAL PRESSURE INDICATOR FORM. This dash number suffix is included in the part number when the filter assembly is to be acquired or used with a specific type of differential pressure indicator. See MIL-F-24402/5 for indicator dash numbers. For logistic support purposes, assemblies will be acquired and stocked without differential pressure indicators. In such cases, indicators will be ordered separately using the applicable part number shown on MIL-F-24402/5.

PART NUMBER CROSS REFERENCE. Table I provides a cross-reference between old (MIL-F-24402D and earlier) and new part numbers.

TABLE I. Cross reference of part numbers.

New part number	Old part numbers	
(Fatigue rated) 1/	(Fatigue rated) 2/	(Non-fatigue rated) 2/
M24402/2NS	M24402-2N-BF-3 Righthand outlet (rev. D and earlier)	M24402-2N-BF-0
M24402/2NL	M24402-2N-BF-3 Lefthand outlet (rev. D and earlier)	M24402-2N-BF-0 Lefthand outlet (configuration not qualified but exists)
M24402/2ND	M24402-2N-BF-3 (rev. D, amendment 1)	
M24402/2RS	M24402-2R-BF-3 1/ Righthand outlet	M24402-2R-BF-0 Righthand outlet
M24402/2RD	M24402-2R-BF-3 (rev. D, amendment 1)	
M24402/2RL		

1/ If a differential pressure indicator is to be acquired with the filter assembly, add the appropriate indicator dash number suffix from the applicable specification sheet.

2/ Add E, G, or M depending on the form of differential pressure indicator acquired with the filter assembly, as follows:

E - Electrical with mechanical pop-up indicator
G - Gauge type indicator
M - Mechanical pop-up indicator

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SUBSTITUTION OF ASSEMBLIES. Assemblies with new part numbers may be substituted for all assemblies with old part numbers as shown in table I. Only assemblies with old part numbers that are fatigue rated may be substituted for assemblies with new part numbers. Unless prevented by the thickness of the blind flange, the fatigue rated assemblies with left and right dual outlet ports may be substituted for either left or righthand outlet port assemblies.

REQUIREMENTS

(a) Configuration. The filter assemblies shall meet the dimensional requirements shown on figure 1.

(1) Type N assemblies. Type N filter assemblies shall be in accordance with the schematic shown on figure 2.

(2) Type R assemblies. Type R filter assemblies shall be in accordance with the schematic shown on figure 3. Type R assemblies shall be similar to type N assemblies except that they shall include a bypass relief valve which will automatically bypass fluid around the filter elements when a specified differential pressure is reached.

(3) Inlet and outlet ports. The inlet and outlet ports of the filter assemblies shall be in accordance with the porting detail dimensions shown on figure 4. For assemblies with two outlet ports, a blind flange in accordance with MIL-F-24704 and MIL-F-24704/4 shall be installed on the specified port utilizing an insert sleeve in accordance with MIL-F-24704 and MIL-F-24704/5.

(4) Filter element size. The filter housings shall utilize size B elements in accordance with MIL-F-24402/4.

(b) Maximum rated operating pressure. The filter assemblies shall have a maximum rated operating pressure of 3000 lb/in^2.

(c) Rated flow capacity. The filter assemblies shall have a rated flow capacity of 100 gallons per minute.

(d) Performance.

(1) Bypass relief requirements. Type R assemblies shall have bypass relief valve settings as specified in table II.

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TABLE II. Bypass relief valve setting.

Test condition	Pressure (lb/in^2)	Test fluid temperature	Leakage rate (mL/min)	
			MIL-L-17331	MIL-H-5606
(a) Cracking pressure				
(1) Qualification	150 minimum	40°C maximum	10 minimum	-
(2) Quality conformance	Not more than 5 lb/in^2 higher than the actual cracking pressure for the qualification assembly	20 to 60°C	10 minimum	50 minimum
(b) Full flow (qualification only)	200 maximum	40°C maximum	N/A	-
(c) Reseat pressure				
(1) Qualification	120 minimum	40°C minimum	4 maximum	-
(2) Quality conformance	120 minimum	20 to 60°C	4 maximum	8 maximum

(2) Fatigue impulse rating. The filter assemblies shall have a fatigue impulse rating of 3000 lb/in^2 in accordance with NFPA T2.6.1.

(3) Proof pressure. The filter assemblies shall withstand a proof pressure of 4500 lb/in^2.

(4) Housing pressure drop. At the maximum rated flow with MIL-L-17331 fluid, the housing pressure drop with minimum restriction dummy elements installed shall not exceed 35 lb/in^2 differential. Selector valve may be positioned to allow flow through both filtration circuits to meet this requirement.

Preparing activity:
Navy - SH
(Project 4330-N057-02)

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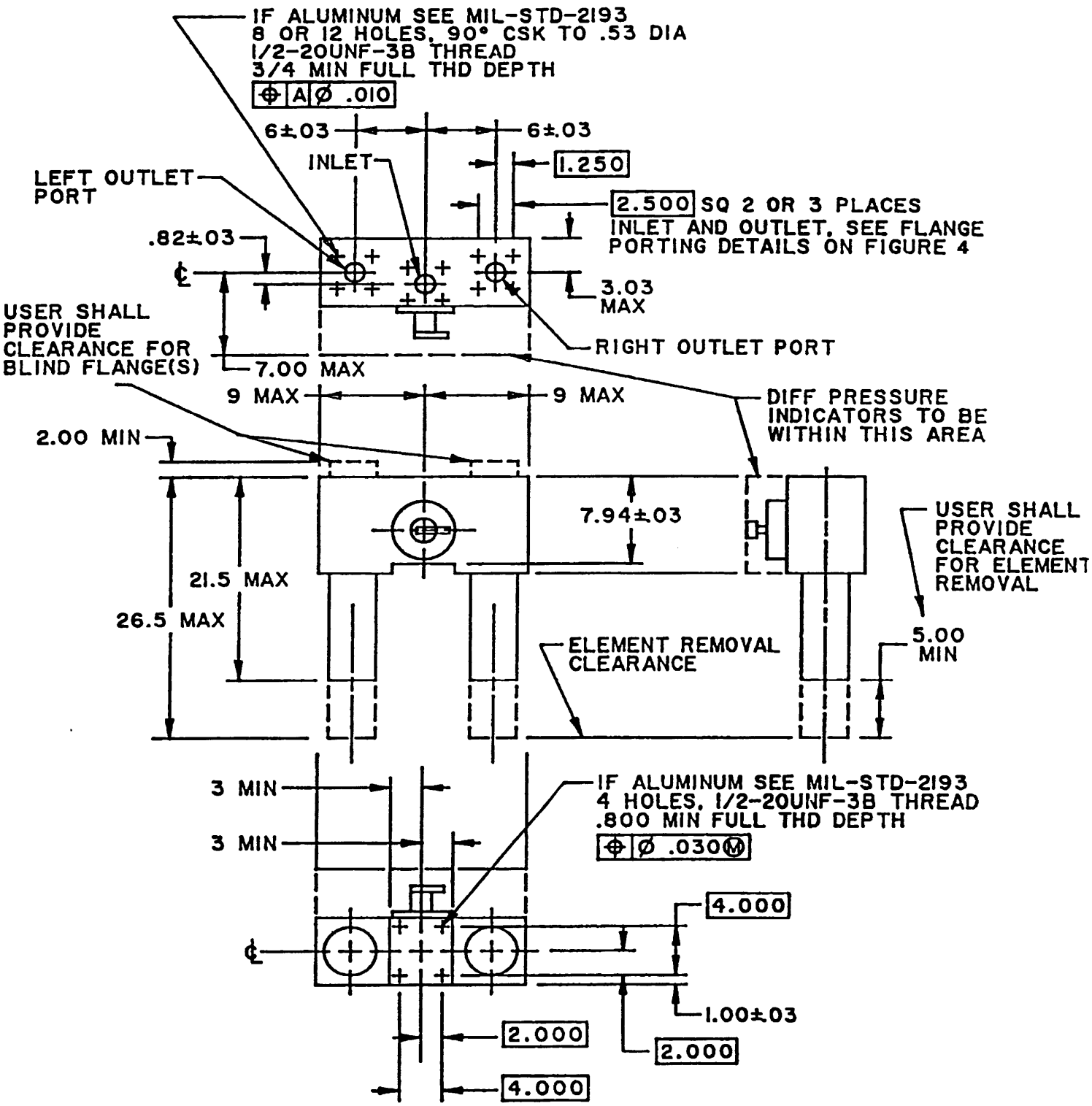


FIGURE 1. Duplex element filter housing outline dimensions.

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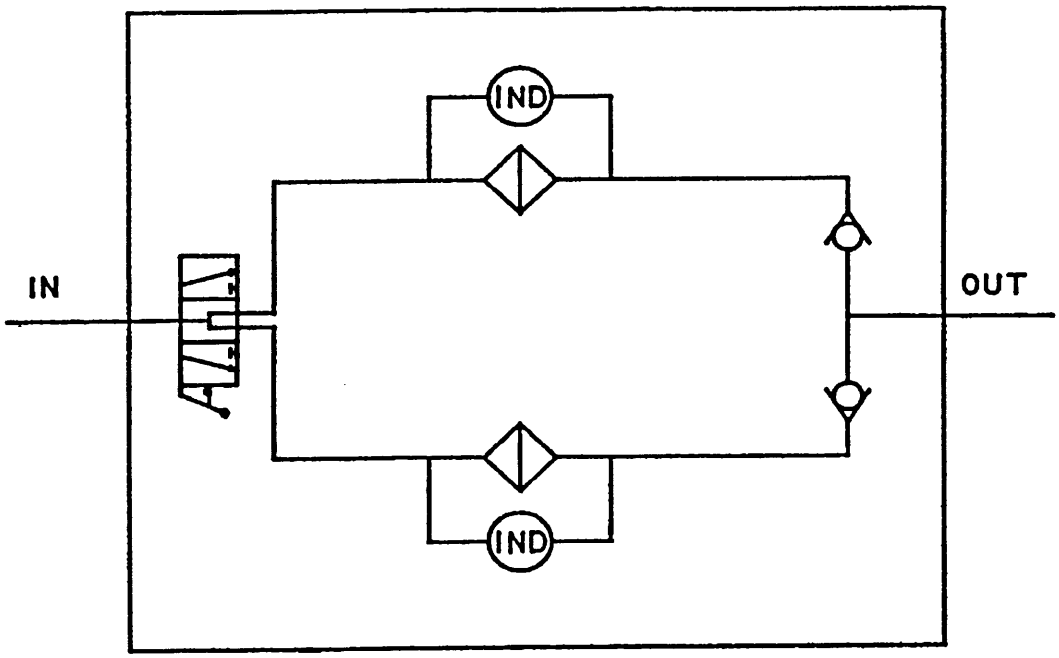


FIGURE 2. Schematic of type "N" filter assembly.

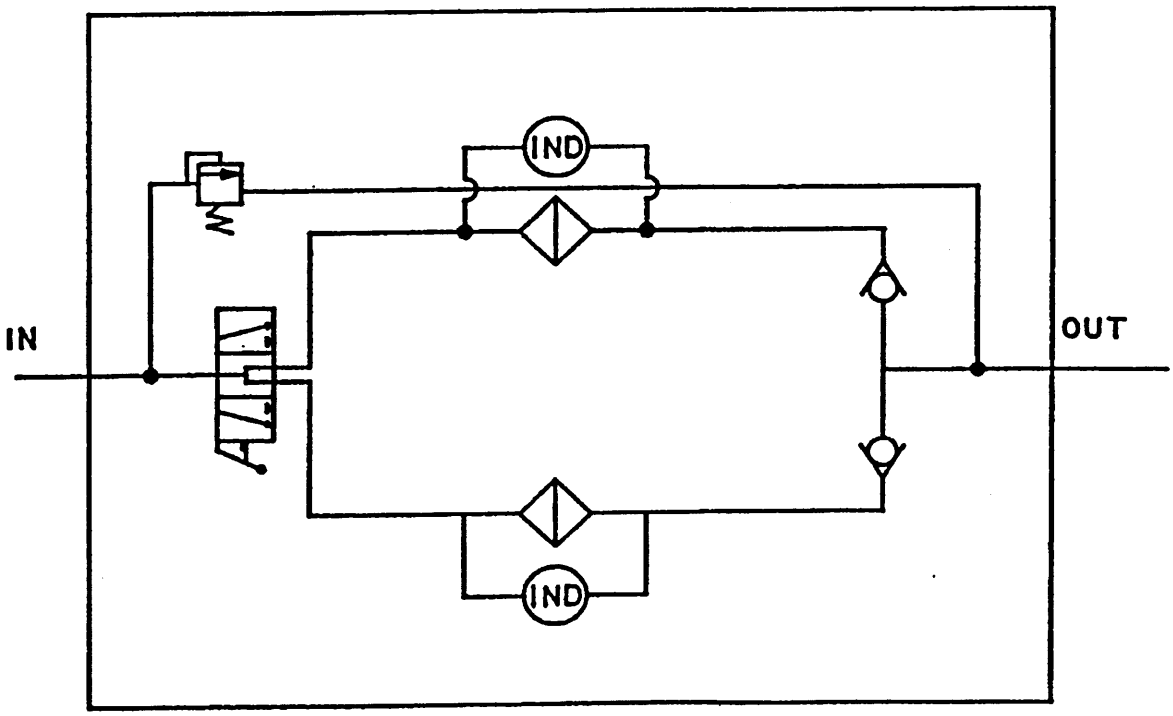


FIGURE 3. Schematic of type "R" filter assembly.

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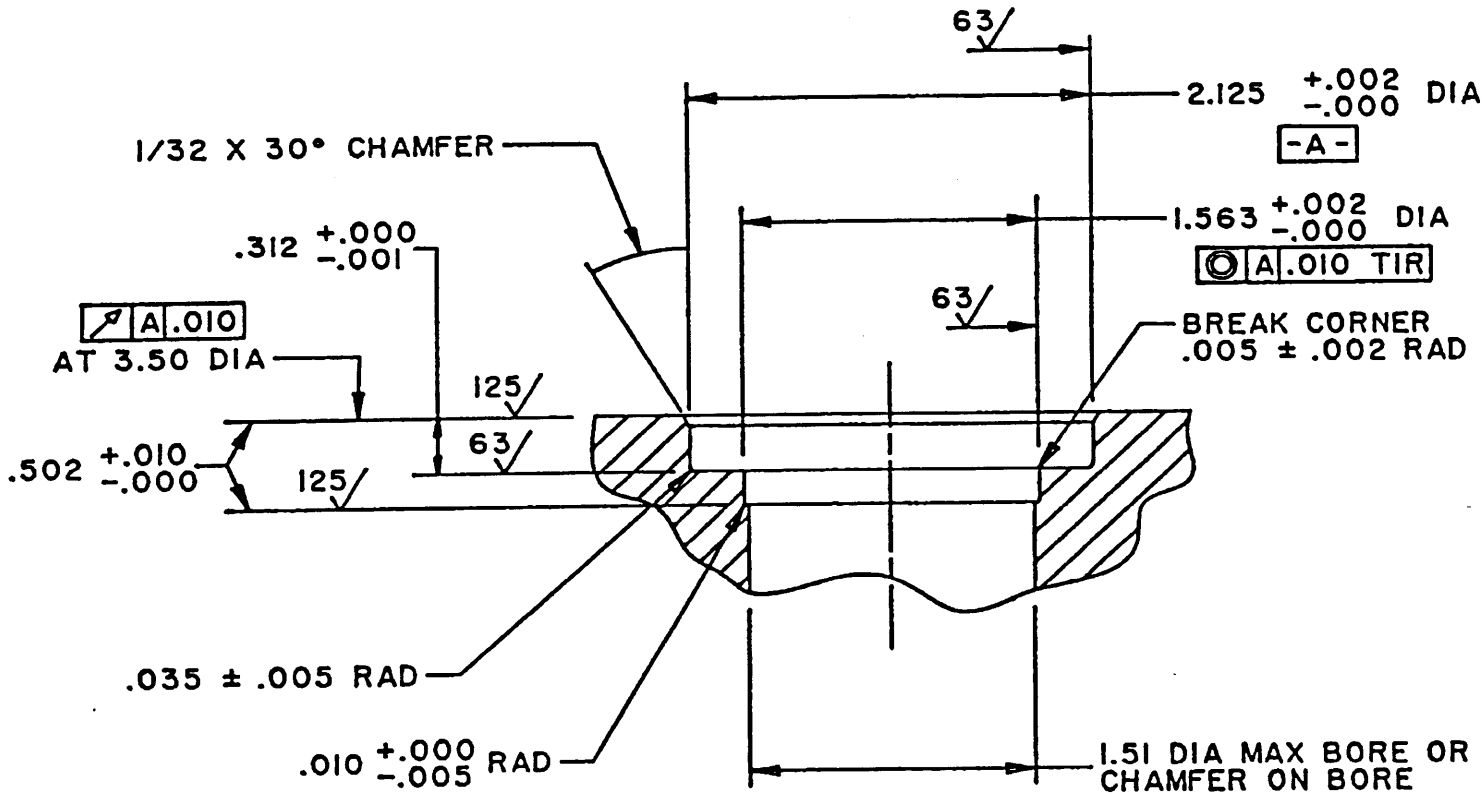


FIGURE 4. Porting details for mating flanges.

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